WEN-HORNG SHEU

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Education

Ph.D. Candidate in Computer Science, University of California, Davis; GPA: 4.0/4.0 2023-Present
· Research Areas: Distributed Algorithms, Streaming Algorithms, Large-Scale Computation.
M.S. in Computer Science, National Tsing Hua University; GPA: 3.9/4.0 2019-2021
· Coursework: Approximation Algorithms, Computational Geometry.
B.S. in Computer Science, National Tsing Hua University; GPA: 3.85/4.0 2015-2019
· Coursework: Introduction to Machine Learning, Parallel Algorithm Design.

Publications

Following the convention in theoretical computer science, author names are ordered alphabetically unless stated otherwise.

7. **Fixed-Threshold Peeling in Sublinear MPC: Round-Approximation Tradeoffs and Applications**
   with Slobodan Mitrović and Theodore Pan
   to appear in DISC 2026 (International Symposium on Distributed Computing)
6. **A Framework for Boosting Matching Approximation: Parallel, Distributed, and Dynamic**
   with Slobodan Mitrović
   SPAA 2025 (ACM Symposium on Parallelism in Algorithms and Architectures)
5. **Faster MPC Algorithms for Approximate Allocation and Matching in Uniformly Sparse Graphs**
   with Jakub Łącki, Slobodan Mitrović, and Srikkanth Ramachandran
   SPAA 2025 (ACM Symposium on Parallelism in Algorithms and Architectures)
4. **Faster Semi-streaming Matchings via Alternating Trees**
   with Slobodan Mitrović, Anish Mukherjee, and Piotr Sankowski
   ICALP 2025 (EATCS International Colloquium on Automata, Languages, and Programming)
3. **Kernelization and Approximation Algorithms for Finding a Perfect Phylogeny from Mixed Tumor Samples**
   Wen-Horng Sheu and Biing-Feng Wang (contribution order)
   TCBB 2025 (IEEE Transactions on Computational Biology and Bioinformatics)
2. **New Algorithms for Constructing Frequency Difference Consensus Trees**
   Biing-Feng Wang, Chih-Yu Li, and Wen-Horng Sheu (contribution order)
   TCBB 2025 (IEEE Transactions on Computational Biology and Bioinformatics)
1. **Parameterized Complexity for Finding a Perfect Phylogeny from Mixed Tumor Samples**
   Wen-Horng Sheu and Biing-Feng Wang (contribution order)
   SIDMA 2023 (SIAM Journal on Discrete Mathematics)

Talks & Presentations

- **A Framework for Boosting Matching Approximation: Parallel, Distributed, and Dynamic**
  INSAIT, May 2025
- **Toward Optimal Semi-streaming Algorithm for $(1+\epsilon)$-Approximate Maximum Matching**
  at the Workshop on Local Algorithms 2024
  Simons Institute for the Theory of Computing, UC Berkeley, Aug 2024

Service & Teaching

Shadow Program Committee Member for ACM SPAA 2026 and 2027

- Invited to serve on the shadow program committee for a leading conference in parallel algorithms.
- Reviewed 6 submissions; participated in committee discussions and contributed to paper evaluations.

External Reviewer for the following conferences and journals.

- Conferences: ICALP (2025), ISAAC (2026), SODA (2025, 2026, 2027), SOSA (2025, 2027), STOC (2026).
- Journals: Distributed Computing (2025), TALG (2025, 2026).

Teaching Assistant at the University of California, Davis 2024-Present

- Courses: Implementation of Data Structures in C (2026), Algorithm Design and Analysis (2024, 2025, 2026), Special Topics in Theoretical Computer Science (2024).

Teaching Assistant at National Tsing Hua University 2019-2022

- Courses: Computational Geometry (2020, 2022), Parallel Algorithm Design (2019, 2022), Design and Analysis of Algorithms (2019, 2020, 2021).

Honors & Awards

Gold Award in *the 2019 ICPC Asia Pacific Taipei-Hsinchu Regional Contest*

- Competed in several team programming contests, including ICPC, as an undergraduate.
- Built strong teamwork and problem-solving skills.

Second Place Award in *the ACM TAU 2018 Contest on Path Reporting*

- Undergraduate research project: created a tool to perform timing analysis on input circuits.

Grandmaster on Codeforces

- Ranked as a grandmaster, within top 1% globally on the online competitive programming platform Codeforces (as of Feb 2021).
- Placed in the top 1% (out of 10,000+ contestants globally) in four online programming contests on Codeforces.

Google Code Jam 2021 Round 3 Qualifier

- Competed in the programming contest Google Code Jam 2021.
- Placed 255th in Round 3, within top 1% of all 37,000+ participants of the qualification rounds.

Research Experience

Graduate Research Assistant at the University of California, Davis 2023-Present

- Research Areas: Distributed Algorithms, Streaming Algorithms.
- Studied the maximum matching problem in distributed and streaming settings.

Research Assistant at National Tsing Hua University 2021-2023

- Research Areas: Parameterized Algorithms, Computational Biology.
- Proposed new algorithms with applications in cancer genomics and phylogenetic analysis.
- Created problems for multiple ICPC regional contests.

Skills

Programming Languages	C++, Python
Tools	Git, LaTeX, LiteLLM, Linux/Bash
Languages	English (fluent, TOEFL 105/120), Chinese (native)
Coursera Certificate	AI Agents and Agentic AI with Python & Generative AI